Nikhil Jain

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I am a Nikhil Jain, an AI Developer by Profession. I have 2.10 years of work experience (1 year in Mechanical Maintenance,1 year in Purchase Executive and 10 months in Data Science, Data Annotat**or** and AI) with B. Tech from Dr.A.P.J. Abdul Kalam Technical University Lucknow. I am having complete blend of mathematics and programming.

Skills:

- Python
- Matplotlib
- Scikit-learn
- Deep Learning
- Machine Learning
- Statistics
- Pandas
- DSA
- NumPy
- Probability
- NLP
- SQL
- SpaCy and NLTK

Notable Achievements:

- Four Times **GATE** examination Qualified
- Cochin Shipyard Limited Kochi Qualified [PSU] (**Enterprises of Central Government**)
- Kelton Trivandrum [PSU] (**Enterprises of State Government**)
- The Fertilizer and Chemical Travancore Limited Kochi Qualified [PSU] (**Enterprises of Central Government**)

EMPLOYMENT SCAN

Innodata India Pvt. Ltd.: Senior Associate – AI & LLM Practice: Oct 2024 - Till Date

- **Project: Video Encoder**

  - **Problem Statement:** The project addresses the need for high-quality video annotations for segmentation and masking tasks, focusing on Delta information observable only through video analysis. Accurate subject tracking and caption generation are critical for enhancing AI model performance.
  - **Objective:** To accurately mask subjects in videos for subsequent captioning to implement effective tracking methods (Exhaustive and Selective) to ensure reliable annotations and to generate precise action captions that describe movements and interactions within the video content.
  - **Data Collections:** I gathered the information needed for this project from datasets provided by the company
  - **Documentations:**
    - Maintained detailed records of annotation work, including any changes or updates to guidelines
    - Provided regular reports on annotation progress and quality metrics to project supervisors
  - **Challenges:**
    - **Subject Identification:** Consistently tracking multiple subjects under varying conditions
    - **Annotation Consistency:** Maintaining high-quality annotations across different videos
    - **Data Quality:** Ensuring captured video data is suitable for effective analysis
  - **Learning:**
    - Understanding video processing techniques and the role of Delta information in action recognition
    - Developing skills in using annotation tools for segmentation and captioning
  - **Business Benefits:**
    - **Higher ROI**: Improved AI accuracy leads to better outcomes, maximizing return on investment
    - **Operational Efficiency:** Streamlined workflows reduce resource expenditure and increase productivity
    - **Enhanced Customer Satisfaction:** Improved user experiences result in higher engagement and loyalty
    - **Competitive Edge:** Access to high-quality data enables clients to leverage advanced AI capabilities, differentiating them in the market.
  - **Roles and Responsibility:**
    - **Video Analysis**: Reviewing content to identify subjects for tracking and masking
    - **Collaboration:** Working with the FAIR Perception team to refine the annotation process
    - **Quality Assurance:** Ensuring data accuracy and consistency
    - **Annotation Execution:** Applying masks and generating action captions

Macgence Technologies Pvt. Ltd.: Data Annotator Intern: April 2024 – Sep2024

- **Project: Named Entity Recognition**

  - **Problem Statement:** Analysis a state-of-the-art Named Entity Recognition (NER) Datasets utilizing Spacy and NLTK libraries to accurately identify and classify named entities in unstructured text data, enabling efficient data processing and informed decision-making across industries.
  - **Objective:** Execute a comprehensive data annotation strategy to deliver high-quality datasets that significantly improve the performance and reliability of NER models used in advanced NLP applications.
  - **Data Collections:** I gathered the information needed to build the NER system from datasets provided by the company.
  - **Tools and Technologies used:**
    - **Annotation Tools:** SpaCy and NLTK libraries
    - **Programming Languages:** Python (for scripting, automation, and data manipulation)
    - **Data Analysis and Management:** Pandas, NumPy, Excel/Google Sheets
    - **Visualization:** Matplotlib, Seaborn
  - **Documentations:**
    - Maintained detailed records of annotation work, including any changes or updates to guidelines.
    - Provided regular reports on annotation progress and quality metrics to project supervisors.
  - **Challenges:**
    - **Data Annotation Complexity:** Overcoming the challenge of accurately labeling diverse named entities in text data, requiring careful consideration of context and consistent annotation guidelines.
    - **Model Optimization:** Optimizing the performance of the NER model to achieve high accuracy and precision, involving experimentation with various techniques such as feature engineering and hyperparameter tuning.

- **Dealing with Ambiguity:** Addressing ambiguity in entity recognition, particularly in cases where named entities have multiple interpretations or appear in varied contexts, necessitating refined algorithms and context-aware processing.

- **Learning:**
  - **Advanced NER Techniques:** Gained expertise in advanced NER techniques, including entity linking and domain-specific entity recognition, to improve the accuracy and effectiveness of the NER system.
  - **Effective Data Handling:** Learned strategies for efficient data preprocessing and annotation, ensuring high-quality training data and robust model performance.
- **Roles and Responsibility:**
  - **Data Annotation**: Accurately annotated text data to train the NER model, ensuring consistent labelling of named entities consider for effective learning.
  - **Data Management:** Accurate labeling, categorizing and tagging of datasets
  - **Quality Assurance:** Ensuring data accuracy and consistency
  - **Technical Skills:** Basic understanding of Python ,SQL and other relevant languages or tools

CodersCave: Data Science Intern: Jan 2024 – March2024

- **Phase 2 Golden Task**

  - **Project Showcase**: Developed a cutting-edge spam filter leveraging advanced NLP techniques and ML algorithms to identify and filter out spam emails effectively
  - **Model**: Implemented a sophisticated neural network architecture comprising text vectorization, embedding layers, bidirectional LSTMs and GRUs, convolutional layers, and dense layers
  - **Performance:** Achieved remarkable precision (99.89%) and recall (99.78%) rates, showcasing the model's exceptional accuracy in classifying spam and non-spam emails
  - **Training:** Trained the model over 8 epochs, incorporating early stopping and model checkpointing techniques to ensure optimal performance and prevent overfitting
  - **Learning:** This project enabled me to deepen my understanding of NLP, ML, and neural networks, enhancing my skills in data preprocessing, model development, and performance optimization.

- **Phase 1 Normal Task**

  - **Project Showcase:** Analyze customer behavior based on transaction data
  - **Project Overview:** Create a project that focuses customer behavior based transactional data
  - **Objective:** Conducting EDA to derive valuable insights on customer preferences, purchasing patterns, and behavior
  - **Key Responsibilities:**

    <> Perform EDA Analysis <> Data Cleaning <> Data Distribution
    <> Data Distribution <> Feature Engineering <> Correlation Analysis
    <> Feature Engineering <> Descriptive Statistics Analysis <> Summary and Key Findings
  - **Challenges & Learnings:**

    Challenges Faced in the Project Identifying Areas for Continued Analysis
    Handling Missing Data, implemented robust data imputation techniques to address gaps
  - **Achievements:** Unearthing the Strategic Insight be the Cornerstone for informing Business Decision Making

Vivo Mobile India Pvt. Ltd.: Purchase Executive: Jan 2022 - April 2023

Rico Fluidtronics Pvt. Ltd.: Senior Technician: Jan 2021 - Jan 2022

Certifications

- **IBM Dara science Professional Certificate (**https://tinyurl.com/v7uzsph2**)**
- **Machine Learning Specialization (** https://tinyurl.com/zsevzmk2**)**
- **Deep Learning Specialization (** https://tinyurl.com/a4wnswd4**)**
- **Natural Language Processing (** http://tinyurl.com/4htdabcc**)**